Software Development Plan

Project Name: `Studify` **Document Version:** `2.0` **Date:** `24/Jul/2026`

Revision History

Date	Version	Description	Author
`11/Jul/2026`	`1.0`	Initial version of the Software Development Plan	`Minh Thư`
`24/Jul/2026`	`2.0`	Revised the Software Development Plan based on the TA's feedback	`Minh Thư`

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1. Introduction

> **Authors:** [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Khánh Linh, Gia Phúc] | **Editor:** [Minh Thư]

1.1 Purpose

The purpose of the Software Development Plan is to describe the development plan for **Studify**, a web application tailored for students and working professionals looking to master conversational and specialized English. The platform aligns its content with the **Common European Framework of Reference for Languages (CEFR)** and customizes learning paths for specific industries such as **IT**.

The project focuses on delivering a robust **Minimum Viable Product (MVP)** featuring adaptive general conversation modules and a specialized curriculum tailored for the **Information Technology (IT)** sector.

The following people are involved in the Software Development Plan:

- Project Manager / Business Analyst / UI-UX: Uses the plan to coordinate project activities, manage project scope, and oversee UI/UX-related work.
- Backend Lead: Uses the plan to coordinate backend development activities and review backend tasks.
- Frontend Lead: Uses the plan to coordinate frontend development activities and review frontend tasks.
- Backend Developer: Uses the plan to understand assigned backend development tasks and their deadlines.
- Frontend Developer: Uses the plan to understand assigned frontend development tasks and their deadlines.

1.2 Scope

This Software Development Plan describes the overall development plan for the **Studify** project.

The project aims to design and develop a dynamic, responsive web application that bridges the gap between traditional language learning and practical, career-oriented communication. The platform merges **CEFR proficiency standards** with data-driven personalization to offer users a structured yet flexible ecosystem for language acquisition.

The primary project phase focuses on delivering a robust **MVP** with:

- Adaptive general conversation modules.
- A specialized curriculum tailored for the **IT** sector.
- AI-driven diagnostic testing upon registration.
- IT-focused modules covering Agile terminologies, client communication, and cross-cultural tech collaboration.
- Scenario-based conversational exercises.
- Progress analytics for daily streaks, vocabulary retention, and CEFR sub-skill milestones.

1.3 Overview

This Software Development Plan contains the following information:

- **Project Overview** — describes the purpose, scope, objectives, key features, and target audience of the Studify project.
- **Project Organization** — describes the project team structure and the roles of the five team members.
- **Management Process** — describes the project schedule, incremental builds, sprint plans, project risks, and future sprint activities.

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2. Project Overview

> **Authors:** [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Khánh Linh, Gia Phúc] | **Editor:** [Minh Thư]

2.1 Project Purpose, Scope, and Objectives

Project Purpose:

The **Studify** project aims to design and develop a dynamic, responsive web application that bridges the gap between traditional language learning and practical, career-oriented communication. By merging **CEFR proficiency standards** with data-driven personalization, the platform will offer users a structured yet flexible ecosystem to accelerate their language acquisition.

The primary phase of the project focuses on delivering a robust **Minimum Viable Product (MVP)** that features adaptive general conversation modules alongside a highly specialized curriculum tailored for the **Information Technology (IT)** sector.

Objectives:

- **Standardized Benchmarking:** Integrate automated diagnostic tools to accurately map user proficiency against CEFR levels (A1 to C2).
- **Contextualized Learning:** Replace generic vocabulary with high-utility, industry-specific language tracks, starting with IT-focused communication such as technical stand-ups, client pitches, and code documentation review.
- **Micro-Learning Architecture:** Deliver bite-sized, interactive lessons optimized for the busy schedules of students and working professionals.
- **Measurable Progression:** Provide transparent dashboard analytics to show skill growth and readiness for real-world professional scenarios.

Project Scope / Key Features:

Module	Functional Description	Target Output
Adaptive Placement	AI-driven diagnostic test upon registration.	Establishes user's baseline CEFR level.
The IT Track	Specialized modules covering Agile terminologies, client communication, and cross-cultural tech collaboration.	Mastery of industry-specific jargon and soft skills.
Interactive Roleplay	Scenario-based conversational exercises, such as simulating a software deployment crisis or a job	Improved verbal fluency and situational confidence.

Module	Functional Description	Target Output
	interview.	
Progress Analytics	Visual data tracking daily streaks, vocabulary retention, and CEFR sub-skill milestones for Speaking, Reading, and Writing.	Actionable insights for targeted improvement.

Target Audience:

- **Tech Students & Graduates:** Individuals looking to pass international language benchmarks or secure internships in multinational corporations.
- **Active IT Professionals:** Software engineers, product managers, and QA leads who need to articulate complex technical concepts clearly to non-technical global stakeholders.

2.2 Assumptions and Constraints

Assumptions:

- The project will be developed by a team of five members consisting of a PM/BA/UI-UX, Backend Lead, Backend Developer, Frontend Lead, and Frontend Developer.
- The team will produce five incremental builds for testing and review.
- The final sprint will include a demo of all project features.
- Cross-functional pairing will be used so that no single team member holds exclusive knowledge of critical areas.
- Centralized and up-to-date project documentation will be maintained to enable other team members to take over responsibilities with minimal disruption.
- If Speech-to-Text or LLM services are unavailable, the application will fall back to standard multiple-choice or text-based assessments.
- Pre-tested, deterministic rule-based evaluation will be used for initial diagnostic tests before relying on AI-generated grading.

Constraints:

- The project is organized into 5 incremental builds for testing and review.
- The final sprint includes a demo of all project features.
- Sprint scope is organized according to the defined sprint schedule.
- A strict Minimum Viable Product (MVP) is focused solely on the IT track and core CEFR proficiency tracking.
- New feature requests must undergo a formal change-impact assessment before approval.
- Feature scope is frozen for each sprint to prevent mid-sprint additions.
- A 20% buffer is allocated in all major sprint milestones to accommodate unexpected delays.
- API monitoring, billing alerts, and daily usage caps are required to prevent unexpected downtime and cost overruns.

2.3 Project Deliverables

Artifact / Deliverable	Phase / Iteration	Target Delivery Date	Audience / Target
Build 1.0	End of Sprint 1 (PA1)	**06/Jun/2026**	Testing and review
Build 2.0	End of Sprint 2 (PA2)	**11/Jul/2026**	Testing and review
Build 3.0	End of Sprint 3 (PA3)	**25/Jul/2026**	Testing and review
Build 4.0 — Release Candidate	End of Sprint 4 (PA4)	**08/Aug/2026**	Testing and review
Build 5.0 — Final Production	End of Sprint 5 (PA5)	**15/Aug/2026**	Final semester demo

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3. Project Organization

> **Authors:** [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Khánh Linh, Gia Phúc] | **Editor:** [Minh Thư]

3.1 Organizational Structure

The project team consists of five members with assigned roles covering project management, business analysis, UI/UX, frontend development, and backend development.

The team structure is:

- **PM/BA/UI-UX**
- **Backend Lead**
- **Backend Developer**
- **Frontend Lead**
- **Frontend Developer**

The Project Plan identifies the following project organization:

Student ID	Name	Email	Role
24127164	Lê Kim Hằng	lkhang2429@clc.fitus.edu.vn	Backend Lead
24127550	Phạm Minh Thư	pmthu2417@clc.fitus.edu.vn	PM/BA/UI-UX
24127510	Nguyễn Kim Thiên Phước	nktphuoc2412@clc.fitus.edu.vn	Frontend Lead
24127217	Hồ Gia Phúc	hgphuc2425@clc.fitus.edu.vn	Frontend Developer
24127198	Nguyễn Khánh Linh	nklinh2421@clc.fitus.edu.vn	Backend Developer

3.2 Roles and Responsibilities

Person	Role	Responsibilities
Phạm Minh Thư	PM/BA/UI-UX	Project management, business analysis, and UI/UX. Responsible for UI/UX design tasks, including interfaces for the Self-Study Dashboard, Pomodoro, and Flashcard features.
Lê Kim Hằng	Backend Lead	Backend leadership. Responsible for developing the core algorithm for the daily study schedule and reviewing backend development tasks.
Nguyễn Kim Thiên Phước	Frontend Lead	Frontend leadership. Responsible for researching and building the Visual Tree Roadmap Component, implementing Selection API functionality for text highlighting and keyboard shortcuts, and reviewing frontend development tasks.
Hồ Gia Phúc	Frontend Developer	Frontend development. Responsible for building lesson and quiz interfaces, implementing the Progress Bar and Daily Schedule Widget, developing Pomodoro countdown logic and audio notifications, and building Flashcard-related UI components.
Nguyễn Khánh Linh	Backend Developer	Backend development. Responsible for developing APIs for lesson tree structures, lesson and quiz content, learning progress updates, and Flashcards.

Missing: The Project Plan does not provide a complete responsibility description for every role across all project disciplines and workflow details.

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4. Management Process

> **Authors:** [Minh Thư] | **Reviewer:** [Thiên Phước, Kim Hằng, Khánh Linh, Gia Phúc] | **Editor:** [Minh Thư]

4.1 Project Estimates

The project is planned across **5 sprints** with a total duration of **10 weeks**:

- **Sprint 1 (PA1):** 3 weeks — Authentication & Onboarding.
- **Sprint 2 (PA2):** 2 weeks — Self-Study Dashboard, Pomodoro.
- **Sprint 3 (PA3):** 2 weeks — Virtual Study Room, Flash Card.
- **Sprint 4 (PA4):** 2 weeks — AI Speaking Assistant.
- **Sprint 5 (PA5):** 1 week — System Testing, Bug Fixing & Final Demo.

The project will produce **five incremental builds**, with each build completed at the end of its corresponding sprint.

A 20% buffer is allocated in all major sprint milestones to accommodate unexpected delays.

4.2 Project Plan

The project follows a sprint-based incremental development plan consisting of 5 sprints and 5 incremental builds.

4.2.1 Phase Plan

Sprint Schedule Overview:

Sprint	Duration	Main Activities	Expected Output
Sprint 1 (PA1)	3 weeks	Authentication & Onboarding	Build 1.0 — Baseline system
Sprint 2 (PA2)	2 weeks	Self-Study Dashboard, Pomodoro	Build 2.0 — Self-Study Dashboard integration and Pomodoro
Sprint 3 (PA3)	2 weeks	Virtual Study Room, Flashcard	Build 3.0 — Virtual Study Room integration and Flashcard
Sprint 4 (PA4)	2 weeks	AI Speaking Assistant	Build 4.0 — Release Candidate
Sprint 5 (PA5)	1 week	System Testing, Bug Fixing & Final Demo	Build 5.0 — Final Production

Build Plan:

- **Build 1.0 (End of Sprint 1):** Baseline system with User Registration, Login, and Onboarding classification flow.
- **Build 2.0 (End of Sprint 2):** Integration of Self-Study Dashboard, visual roadmap, lesson/quiz modules, and Pomodoro widgets.
- **Build 3.0 (End of Sprint 3):** Addition of the Virtual Study Room including RBAC, file sharing, real-time chat via WebSocket, and Flashcard feature.
- **Build 4.0 (End of Sprint 4 — Release Candidate):** Full system integration including the AI Speaking Assistant with STT & LLM processing.
- **Build 5.0 (End of Sprint 5 — Final Production):** Fully tested, bug-free build ready for the final semester demo.

4.2.2 Iteration Objectives

Sprint-based objectives available in the Project Plan:

- **Sprint 1:** Authentication & Onboarding.
- **Sprint 2:** Self-Study Dashboard, Pomodoro.
- **Sprint 3:** Virtual Study Room, Flashcard.
- **Sprint 4:** AI Speaking Assistant.
- **Sprint 5:** System Testing, Bug Fixing & Final Demo.

4.2.3 Releases

The project will produce five incremental builds:

- **Build 1.0:** Baseline system with User Registration, Login, and Onboarding classification flow.
- **Build 2.0:** Self-Study Dashboard, visual roadmap, lesson/quiz modules, and Pomodoro widgets.

- **Build 3.0:** Virtual Study Room with RBAC, file sharing, real-time chat via WebSocket, and Flashcard feature.
- **Build 4.0 — Release Candidate:** Full system integration including the AI Speaking Assistant with STT & LLM processing.
- **Build 5.0 — Final Production:** Fully tested, bug-free build ready for the final semester demo.

4.2.4 Project Schedule

Build Plan:

Phase / Milestone	Target Start Date	Target Completion Date	Milestone / Release Type
Sprint 1 (PA1)	**23/May/2026**	**06/Jun/2026**	Authentication & Onboarding
Sprint 2 (PA2)	**06/Jun/2026**	**11/Jul/2026**	Self-Study Dashboard & Utilities
Sprint 3 (PA3)	**11/Jul/2026**	**25/Jul/2026**	Virtual Study Room, Flashcard
Sprint 4 (PA4)	**25/Jul/2026**	**08/Aug/2026**	AI Speaking Assistant
Sprint 5 (PA5)	**08/Aug/2026**	**15/Aug/2026**	System Testing, Bug Fixing & Final Demo
Build 1.0	**23/May/2026**	**06/Jun/2026**	Baseline system
Build 2.0	**06/Jun/2026**	**11/Jul/2026**	Self-Study Dashboard integration
Build 3.0	**11/Jul/2026**	**25/Jul/2026**	Virtual Study Room integration
Build 4.0	**25/Jul/2026**	**08/Aug/2026**	Release Candidate
Build 5.0	**08/Aug/2026**	**15/Aug/2026**	Final Production

Detailed Plan for Upcoming Sprint (Sprint 2):

- **Focus:** Virtual Study Room & Flashcard Enhancement
- **Duration:** 2 Weeks (e.g., July 11, 2026 - July 25, 2026)

Task Description	Assignee (Doer)	Reviewer	Due Date
[UI/UX] Design UI-UX interfaces for Virtual Study Room, including Group Management, Task Widget, Document Repository, real-time Chat Box, and Flashcard features, with detailed user flows and button functionalities.	Minh Thư (PM/BA/UI-UX)	Thiên Phước (Frontend Lead), Kim Hằng (Backend Lead)	13/Jul/2026
[UI/UX] Finalize the UI/UX design for Group Management, including Create Group, Join Group via Code, Group Member List, and Group Leader controls.	Minh Thư (PM/BA/UI-UX)	Thiên Phước (Frontend Lead)	14/Jul/2026

Task Description	Assignee (Doer)	Reviewer	Due Date
[BE] Implement Role-Based Access Control (RBAC) for Group Leader and Group Member roles, including permission validation for group-related actions.	Kim Hằng (Backend Lead)	Minh Thư (PM/BA/UI-UX)	16/Jul/2026
[BE] Develop a unique and secure Group Code generation algorithm for creating and joining study groups.	Kim Hằng (Backend Lead)	Khánh Linh (Backend Dev)	16/Jul/2026
[BE] Build APIs for Create Group, Join Group via Code, Leave Group, and Retrieve Group Information and Member List.	Khánh Linh (Backend Dev)	Kim Hằng (Backend Lead)	18/Jul/2026
[FE] Build the Group Management interface, including Create Group, Join Group via Code, Group Details, and Member List components.	Thiên Phước (Frontend Lead)	Minh Thư (PM/BA/UI-UX)	19/Jul/2026
[BE] Design database schema and develop APIs for Task Assignment, including task creation, assignment to group members, status updates, and task retrieval.	Khánh Linh (Backend Dev)	Kim Hằng (Backend Lead)	20/Jul/2026
[FE] Build the Task Widget and Task Assignment interface, supporting task creation, member assignment, task status tracking, and task completion.	Gia Phúc (Frontend Dev)	Thiên Phước (Frontend Lead)	21/Jul/2026
[BE] Configure AWS S3/Cloudinary storage and develop Upload/Download APIs for study materials and group documents.	Kim Hằng (Backend Lead)	Khánh Linh (Backend Dev)	21/Jul/2026
[FE] Build the Document Repository interface with file drag-and-drop upload, file list display, download, and basic file management functionalities.	Gia Phúc (Frontend Dev)	Thiên Phước (Frontend Lead)	22/Jul/2026
[BE] Set up the WebSocket (Socket.io) architecture and required events for real-time group chat communication.	Kim Hằng (Backend Lead)	Thiên Phước (Frontend Lead)	22/Jul/2026
[FE] Connect the WebSocket client and implement real-time Chat Box functionality, including message sending, receiving, and chat state management.	Thiên Phước (Frontend Lead)	Kim Hằng (Backend Lead)	23/Jul/2026
[FE] Implement the Flashcard study interface and interaction logic, including card flipping, keyboard shortcuts, and vocabulary review flow.	Gia Phúc (Frontend Dev)	Thiên Phước (Frontend Lead)	23/Jul/2026

Task Description	Assignee (Doer)	Reviewer	Due Date
[BE] Integrate Flashcard data with the existing Flashcard APIs and ensure synchronization between Flashcard collections, Tags, and the study interface.	Khánh Linh (Backend Dev)	Kim Hằng (Backend Lead)	24/Jul/2026
[Documentation] Prepare the Weekly Report and AI Usage Report documenting the team's progress, completed tasks, AI-assisted activities, and lessons learned during the sprint.	Minh Thư (PM/BA/UI-UX)	All Team Members	25/Jul/2026

High-Level Plan for Future Sprints:

Sprint 4: AI Speaking Assistant:

- **Self-Training & R&D:** Research STT APIs, Web Audio API, and LLM Prompt Engineering.
- **Design:** UI for AI Voice Chat interface and detailed evaluation display.
- **Backend:** * Integrate third-party STT API.
- Design and test Prompt Engineering for LLM (GPT-4o/Claude) to output strict JSON (Grammar, Vocab, Relevance, Suggestions).
- Build APIs to receive audio streams, process via AI, and save history/scores.
- **Frontend:** * Implement Web Audio API/MediaRecorder for precise voice recording.
- Build Chat UI with loading animations and render JSON evaluation data.
- **Documentation:** Weekly Report and AI Usage Report for the recent Sprint.

Sprint 5: Testing, Bug Fixing & Demo Preparation:

- **Testing:** Conduct End-to-End (E2E) user flow testing across all modules.
- **Bug Fixing:** Resolve issues identified in UI, Backend logic, and Real-time connections.
- **Optimization:** Refactor code and optimize Database queries.
- **Deployment:** Finalize production environment deployment.
- **Documentation & Demo:** Prepare final presentation slides, rehearse the product demo, and finalize the PA5 Project Report.

4.2.5 Project Resourcing

The project requires five team members with the following roles:

1 PM/BA/UI-UX: Phạm Minh Thư. 1 Backend Lead: Lê Kim Hằng. 1 Backend Developer: Nguyễn Khánh Linh. 1 Frontend Lead: Nguyễn Kim Thiên Phước. 1 Frontend Developer: Hồ Gia Phúc.

The project requires skills and experience in:

- Project management and business analysis.
- UI/UX design.
- Frontend development.
- Backend development.
- Database schema design.

- API development.
- WebSocket real-time communication.
- AWS S3/Cloudinary file storage.
- Speech-to-Text APIs.
- Web Audio API and MediaRecorder.
- LLM Prompt Engineering.
- Testing, bug fixing, database optimization, and deployment.

4.3 Project Monitoring and Control

Project monitoring and control are performed through:

- Sprint Schedule Monitoring: The project is divided into five sprints with defined durations and objectives.
- Task Assignment: Tasks are assigned to specific team members.
- Task Review: Each Sprint 2 task has an assigned reviewer.
- Due Date Tracking: Tasks have defined due dates.
- Incremental Build Monitoring: Five incremental builds are produced for testing and review.
- Risk Management: Project risks are identified and mitigation strategies are defined.
- Scope Control: New feature requests require a formal change-impact assessment, and sprint scope is frozen to prevent mid-sprint additions.
- Documentation: Weekly Reports, AI Usage Reports, final presentation slides, and the PA5 Project Report are included in the project plan.
- Testing and Quality Control: Sprint 5 includes End-to-End user flow testing, bug fixing, code refactoring, and database query optimization.
- Deployment Monitoring: The production environment is finalized during Sprint 5.

4.3.1 Requirements Management

The Project Plan defines the project scope through the MVP, key features, sprint schedule, build plan, and high-level plans for future sprints.

Scope control is addressed through the following risk mitigation strategies:

- Define a strict **MVP** focused solely on the IT track and core CEFR proficiency tracking.
- Maintain a dedicated **"Nice-to-Have" backlog** for Phase 2 features.
- Require all new feature requests to undergo a formal **change-impact assessment** before approval.
- Freeze the feature scope for each sprint to prevent mid-sprint additions.

4.3.2 Reporting and Measurement

Sprint tasks are monitored using:

- Task Description.
- Assignee (Doer).
- Reviewer.
- Due Date.

Project progress is also evaluated through the completion of the five incremental builds and the testing and review activities associated with each build. Sprint 5 includes End-to-End user flow testing across all modules, bug fixing, code refactoring, database query optimization, and production environment deployment.

4.3.3 Risk Management

The Project Plan identifies the following risks:

Risk Ranking	Risk Description and Impact	Mitigation Strategy and/or Contingency Plan
High	**Scope Creep ("Just One More Feature" Trap)** — Impact: High	Define a strict MVP focused solely on the IT track and core CEFR proficiency tracking. Maintain a dedicated "Nice-to-Have" backlog for Phase 2 features. Require all new feature requests to undergo a formal change-impact assessment before approval. Freeze the feature scope for each sprint to prevent mid-sprint additions.
High	**API Failures or Inaccurate CEFR Grading** — Impact: High	Design a graceful degradation mechanism so that if Speech-to-Text or LLM services are unavailable, the application falls back to standard multiple-choice or text-based assessments. Use pre-tested, deterministic rule-based evaluation for initial diagnostic tests before relying on AI-generated grading. Implement API monitoring, billing alerts, and daily usage caps to prevent unexpected downtime and cost overruns. Regularly validate AI-generated CEFR scores against human-reviewed samples to ensure grading accuracy and consistency.
Medium	**Member Unavailability & Bottlenecks** — Impact: Medium	Establish cross-functional pairing so no single team member holds exclusive knowledge of critical areas. Allocate a 20% buffer in all major sprint milestones to accommodate unexpected delays. Maintain centralized and up-to-date project documentation, enabling other team members to take over responsibilities with minimal disruption.

4.3.4 Configuration Management

Configuration and scope control for the project include:

- Maintaining centralized and up-to-date project documentation.
- Performing a formal change-impact assessment for new feature requests before approval.
- Freezing the feature scope for each sprint to prevent mid-sprint additions.
- Maintaining a dedicated "Nice-to-Have" backlog for Phase 2 features.
- Including project documentation and deliverables as part of the planned project outputs.

The project also includes final documentation and deliverables such as Weekly Reports, AI Usage Reports.